

S4 First Internal Examination, July 2023

19-205-0404: APPLIED THERMODYNAMICS

Time : 2 Hours

Maximum Marks : 40

Course Outcomes:

1. Learn first and second law of thermodynamics and the application of these laws for various thermodynamic processes.
2. Apply the knowledge of thermodynamic laws to various engineering systems.
3. Analyse the performance of steam generators with additional accessories used for enhancing the performance.
4. Understand the various types of nozzles and its performance at various back pressure conditions and also for different flow conditions.
5. Calculate the mole and mass fractions of various mixtures and also various specific properties of the mixtures.

Use separate answer books for Module 1 and 4

Module - 4		Marks	BL	CO
1	Derive an expression for the mean value of gas constant of a mixture of ideal gases.	4	2	5
2	What is equivalent molecular mass? Derive an expression for that.	4	2	5
3	Differentiate between ideal gas, perfect gas and real gas	3	1	5
4	State Dalton's law and Amagat's law of mixture of ideal gases	3	1	5
5	The volumetric composition of a sample gas is as follows: H ₂ 50%, CH ₄ 20%, CO ₂ 25% and N ₂ is 5%. Convert it to gravimetric analysis. Find out the mean value of gas constant and apparent molecular mass of the mixture.	6	4	5
Module - 1		Marks	BL	CO
6	Prove the equivalence of the second law of thermodynamics.	4	2	1
7	State and prove Carnot's theorem.	6	2	1
8	When water at 10 MPa and 200°C expands through a throttle to a final pressure of 0.5 MPa, the enthalpy does not change because the throttle is assumed to be adiabatic and no shaft work is done. What is the final state of the water?	5	4	1
9	Explain the concept of Clausius inequality.	5	3	1
